

Self-humidifying effect of air self-circulation system for proton exchange membrane fuel cell engines

Qinguo Zhang ^{a, b, c, d}, Zheming Tong ^{a, b, c, *}, Shuiguang Tong ^{a, b, c}, Zhewu Cheng ^{a, b, c, d}

^a Ningbo Research Institute, Zhejiang University, Ningbo, 315100, China

^b State Key Laboratory of Fluid Power and Mechatronic Systems, Zhejiang University, Hangzhou, 310027, China

^c School of Mechanical Engineering, Zhejiang University, Hangzhou, 310013, China

^d Ningbo Institute of Technology, Zhejiang University, Ningbo, 315100, China



ARTICLE INFO

Article history:

Received 20 June 2020

Received in revised form

17 October 2020

Accepted 22 October 2020

Available online 27 October 2020

Keywords:

Proton exchange membrane fuel cell (PEMFC)

Cathode exhaust circulation

Self-humidifying effect

Liquid water saturation

ABSTRACT

A dynamic mechanism model of a control-oriented hydrogen fuel cell system based on the cathode exhaust gas cycle is established, and the effects of operating parameters on the two-phase water transfer and dynamic output performance of a fuel cell are analyzed, the transport process of material and the spatial distribution of water content in the membrane electrode were obtained. The study found that too low humidity at the cathode inlet of the stack will cause corrosion of the catalyst layer and the carbon black carrier, resulting in agglomeration and sintering of the catalyst particles. The membrane is also severely damaged by mechanical properties, and black impurities and broken fibers are found on the membrane surface. Under the passive operation of the cathode self-circulation system, the maximum humidity of the mixed gas can reach 60%. The new system is advantageous to the evaporation of water vapor at higher operating temperature and can inhibit the flooding of membrane electrode assembly. It takes about 20s for the current to reach the steady state of the set value, and about 30s for the voltage to reach the steady state. The voltage fluctuation is small and the output power is stable, indicating high reliability.

© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

With the increasing number of vehicles in the world and the increasingly serious energy problems, energy saving and environmental protection have become important issues in the development of the vehicle field [1,2]. Proton exchange membrane fuel cell (PEMFC) with green and high efficiency advantages is an energy conversion device that can directly convert chemical energy stored in fuel into electric energy, which is becoming one of the main development directions of new energy vehicles [3]. At present, the durability of automotive fuel cells is one of the technical challenges that limit their commercialization, especially the service life of membrane electrodes [4]. In the working process of the fuel cell, the main role of the membrane electrode assembly is to promote the occurrence of electrochemical reaction processes, transport electrons and protons, and is the most important component [5].

Since the perfluorosulfonic acid membrane widely used in proton exchange membrane fuel cells can conduct hydrogen ions only at certain humidity, too low humidity will cause the membrane's proton conductivity to be greatly reduced, resulting in increased internal resistance and decreased output performance. Furthermore, it leads to more serious membrane dry failure, even membrane tearing, which seriously affects the output performance and remaining life. However, if there is too much moisture in the humidified gas, the electrode will be flooded, which will increase the resistance of the gas entering the electrode, and the performance of the fuel cell will decrease [6]. Therefore, reasonable control of the intake air humidity is important for the efficiency and performance of the fuel cell system [7]. One of the biggest problems in the popularization of fuel cell vehicles is cost reduction. Humidifiers are one of the high-cost components of PEMFC engines.

Accurately revealing the mechanism of PEMFC water flooding and membrane dry failure, so as to effectively diagnose the state of water flooding and membrane dry failure, and to find solutions after the fault has occurred, has gradually become a hot spot in PEMFC research [8]. Among the many mechanisms of flooding and dry failure, Pei [9] simulated the water transfer mechanism of

* Corresponding author. State Key Laboratory of Fluid Power and Mechatronic Systems, Zhejiang University, Hangzhou, 310027, China.

E-mail address: tzm@zju.edu.cn (Z. Tong).

PEMFC; it is found that the existence of gas diffusion layer, gas diffusion layer and catalyst layer microporous layer with large porosity can prevent water droplets from blocking pores through larger diameter water droplets, which can effectively avoid flooding failure. Song [10] analyzed the channel contact angle and the channel material of PEMFC by numerical simulation and found that when the channel contact angle was 90° , a layer of water film would be formed on the wall of the channel, the occurrence of a flood fault. In the experimental studies of Ropak [11,12], it was found that the electric drag is too strong when the current density is large, which causes water shortage or film failure on the anode side, and the ohmic resistance exceeds $400 \text{ m}\Omega \text{ cm}^2$. Under the conditions of high current density and high relative humidity, excess water cannot be discharged in time because it blocks the pores of the flow channel and the diffusion layer, resulting in flooding failure. Xu [13] used water invasion-evaporation experiments to simulate the effect of dry-wet cycle on the internal structure of PEMFC and showed that the water flooding-membrane dry cycle would cause the catalyst aggregate particles to increase sharply, accompanied by irreversible damage such as pinholes and catalyst tearing, leading to long-term performance degradation. The above-mentioned mechanism studies show that water flooding and membrane dry failures have severe irreversible damage to PEMFC performance and internal structure, but can be avoided or alleviated from water flooding and membrane dry failures from the materials, internal structure, and operating conditions. External conditions will affect the internal water content of the PEMFC, which in turn affects performance during operation, such as current, temperature, intake pressure, flow, and humidity. Eun [14] proposed that improper operating conditions of the fuel cell would break the water balance of the stack, leading to water flooding and membrane failure. Through experimental tests, it was found that the temperature has a greater impact on the internal humidity of the PEMFC than the stoichiometric ratio of the cathode gas. Chen [15] proposed that too high humidity, too low gas flow rate, and too much internal pressure in the stack could cause uneven distribution of reactants and increase gas flow resistance, cause corrosion of the gas diffusion electrode and blockage of the flow path, and eventually cause flooding failure. The above analysis shows that when the thermal management and water management are out of balance due to improper parameter setting during operation, the liquid water accumulation in the reactor cannot be discharged or the proton exchange membrane contains too little water, which affects the proton conduction, so PEMFC enters into the unhealthy operation state of water flooding and membrane drying.

In order to better observe the change rule of water in fuel cell for better water management, more and more researchers use computational fluid dynamics methods to study the internal reaction mechanism of fuel cell. Dadda [16] studied the water and heat transfer process of proton exchange membrane by using two-dimensional and transient non isothermal model. It was found that when the water content at the cathode side of the membrane was low, the voltage of PEMFC was significantly reduced, and the reverse diffusion of water in the membrane would lead to the long-term water shortage and irreversible damage of the membrane. Peng [17] studied the influence of hydrophobic characteristics of airway wall on the accumulation of liquid water in the channel. Liu [18] simulated the accumulation process and movement rule of droplets in the airway, and found that the droplets in the airway would merge to form a plunger flow. Kim [19] developed a two-phase flow model of three-dimensional complex flow field, and found that the Forchheimer effect enhanced the liquid water in the flow field, and added additional flow resistance around the baffle, thus improving the liquid water and mass transfer at the interface. Cao [20] used a three-dimensional two-phase non-isothermal

model to study the interaction between water and heat transport processes, and found that ribs and channel widths have a significant effect on heat and mass transfer processes. Li [21] simulated the influence of liquid water distribution on reactant transport and current density distribution under different working conditions, and showed that the inlet flow rate and wall wettability had a significant impact on the transport process and distribution of liquid water, and then affected the local reactant transport, and had a considerable interference on current density. Dinah [22] used three-dimensional CFD method to simulate the generation and transportation of water in PEMFC. Yana [23] studied theoretically the dynamic process of water transport in PEMFC, and showed that the influence of water content in PEM and time humidity and thickness of PEM to reach stable state in the process of start-up. When PEM is thinner, the water distribution is more uniform, the water content is higher, and the time to reach stable state is obviously shortened. Chen [24] found that higher initial current density; higher water flow on anode side and smaller operating current can achieve water balance in a shorter time, so as to obtain better dynamic performance of fuel cell.

In order to improve the proton conductivity of the membrane, an external humidifier is usually used to humidify the air, but the external humidifier increases the volume and cost of the system. In addition, an important problem is that the residual liquid water inside the external humidifier will freeze in an environment below zero, and its heat capacity is relatively large, which affects the cold start speed of the fuel cell. Therefore, removing the external humidifier becomes important research directions. Buechi [25] found that PEMFC can run stably for 1800 h without humidification, but the output performance of the reactor decreased by 20–40% compared to the performance under full humidification. William [26] carried out experiments on fuel cells using hydrophobic gas diffusion layers, and the results showed that the stack performance was reduced by 17% compared to full humidification. Qi [27] designed a dual-inlet and dual-outlet dual-channel flow path so that the dry intake air can be fully humidified during the flow process. In addition, the self-humidification of PEMFC by adding hydrophilic porous polyester fiber and platinum and absorbent materials such as SiO_2 and TiO_2 into the proton exchange membrane, but the performance of the above method was lower than that of external humidifier [28–30]. The self-humidification of the stack based on the cathode recirculation system has comprehensive advantages in terms of performance and cost. Kim [31] tested the exhaust gas recirculation cathode humidification system in five fuel cells low pressure systems and found that increasing the recirculation ratio increased intake air humidity and reduced O_2 concentration. Shao [32] simulation found that compared with anode recirculation, the humidification effect of the cathode recirculation using a circulation pump is much better, especially at low relative humidity. However, it has also been found that the power consumption of the cathode recirculation pump is much higher than that of the anode recirculation [33]. Rodosik [34] found that high to medium recirculation ratios were able to stabilize and homogenize the cells voltages along the stack. Besides, large relative humidity ranges were achieved at stack inlet, which can vary from 25 to 85%. Zhao [35] experiment found that the stack cathode inlet can always have a higher RH with the double-circulation system of the circulation pump. Hu [36] studied the control strategy of water content in the stack based on air recirculation under the condition of dry membrane and flooded membrane by using a fuel cell system simulation model. Although the recirculation system using a circulation pump can achieve high humidification, the power consumption of the circulation pump is large, which reduces the output power of the engine, so it is not suitable for use in vehicles.

In practical application, PEMFC system is a complex system with

strong nonlinearity and high coupling among subsystems, which is the main difficulty of the whole system control design. In view of the above deficiencies, a one-dimensional simplified stack model along the proton conduction direction is established to describe the concentration distribution of reaction gas, water vapor and liquid water and their influence on each other. The study aims at establishing an accurate model of the PEMFC engine, which provides with true results of modeling and simulation of self-humidification system with air self-circulation mode. In this work, the system was equipped with a 100-cell stack used to verify the validity of the simulation model and the negative impact of low humidity on MEA. We studied the impact of self-circulating humidification system under various current densities, temperatures, stoichiometric ratios and recirculation ratios, and a comparison with passive humidification was done. Based on PEMFC model simulation with exhaust gas circulating cathode, the results of this paper can provide a theoretical basis for high-performance, long-life design of PEMFC and can provide more specific guidance for fuel cell control optimization.

2. Model development

2.1. PEMFC engine with anode and cathode double cycle

The fuel cell system for an automobile is a clean and efficient energy device, which is mainly composed of hydrogen system, air system, cooling system and control system, as shown in Fig. 1. When hydrogen is pumped into the fuel cell's anode, electrons are released at the anode and transferred to the cathode via an external circuit. While ions are transferred to the cathode through the electrolyte for electrochemical reaction with oxygen to generate a large amount of water, thus forming a closed circuit to generate

electric flow. At the same time, because of the fuel's own reaction and the internal resistance of the fuel cell, the fuel cell must also emit a certain amount of heat to maintain its constant operating temperature. Since the thickness of the proton exchange membrane is only a few microns, the anode recycling can have a certain degree of self-humidification function, but throwing it at the cathode inlet will cause the problem of membrane drying. This paper designs a self-humidifying structure that utilizes self-circulation of air to replace the conventional humidifier, which reduces the volume and cost of the system. An electronic three-way valve is installed at the outlet of the stack to control part of the high-temperature and high-humidity air to self-circulate to achieve the purpose of humidifying the cathode. In addition, the method of self-circulation of air is adopted, and the exhaust gas directly enters behind the filter, which increases the flow rate of the intake air, thereby increasing the intake air volume and reducing the power consumption of the air compressor.

2.2. PEMFC system model

2.2.1. Model hypothesis

PEMFC engine is a complex system of “gas-water-electric-thermal” multiple physics coupling, which is mainly solved by partial differential equations. Under the premise of ensuring that the main reaction of fuel cell is fully described, the following hypotheses are made to simplify the model establishment and calculation in this paper [37]:

- (1) The reaction gas is ideal and incompressible, and the fluid flows in the flow channel as laminar flow.
- (2) The temperature of the stack is constant, without considering the thermal effects caused by temperature gradients.

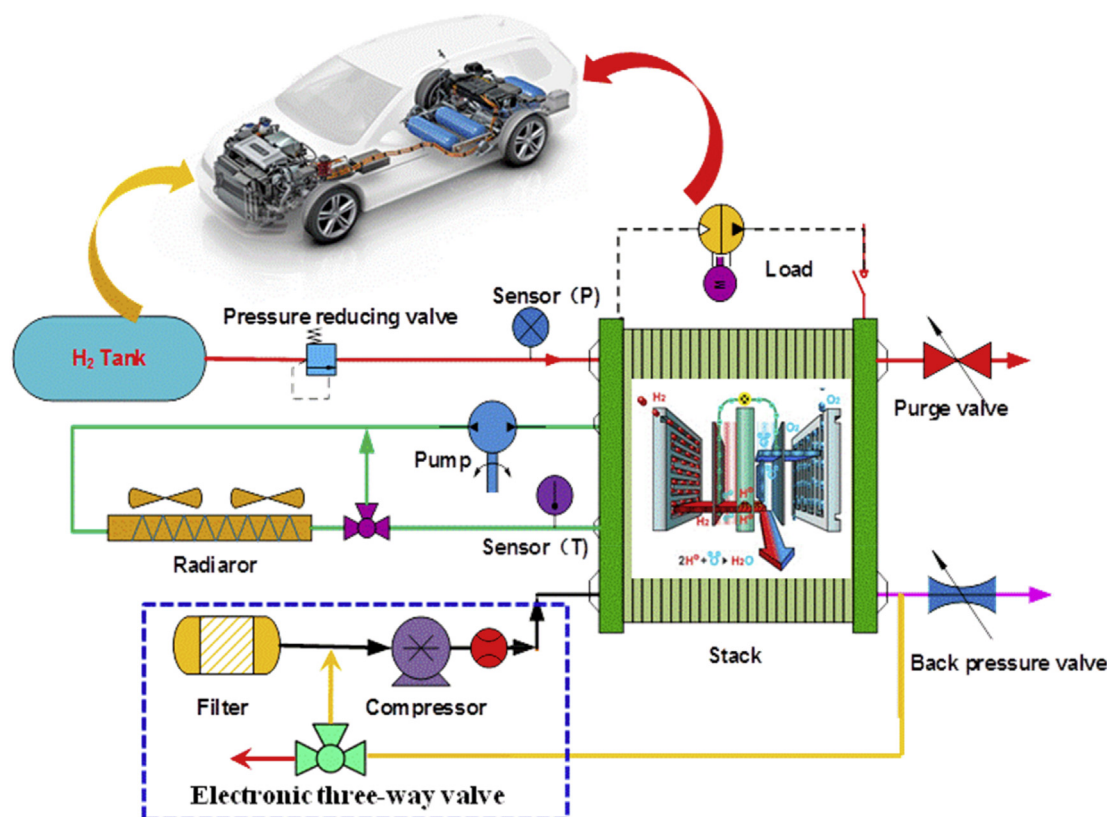


Fig. 1. Fuel cell engine with cathode and anode double circulation system.

- (3) The porous layer of the diffusion layer, catalyst layer and proton exchange membrane is isotropic.
- (4) Only the contact resistance of the contact surface between the current collecting plate and the gas diffusion layer is considered.
- (5) When the partial pressure of water vapor in the gas mixture exceeds the saturation pressure at the local temperature, the water vapor will condense into liquid water.

2.3. Governing equations

2.3.1. Regulated valve with intermittent purging

When the regulated valve with intermittent purging is energized, electromagnetic force is generated to control the opening and closing of the valve port, thereby generating a certain flow and pressure at the output of the solenoid valve, and the structural parameters of the solenoid valve are shown in Table 1.

The equation of pressure flow through the fixed orifice is:

$$q_R = \left[\frac{d_R^2 \cdot A_R^3}{196 \cdot \rho^2 \cdot \nu \cdot l_R} \right]^{1/3} \cdot (p_{R1} - p_{R2})^{2/3} \quad (1)$$

where q_R is the flow through the damping hole; d_R is the diameter of the damping hole; A_R is the flow area of the damping hole; l_R is the length of the damping hole; ρ is the oil density; ν is the oil viscosity; p_{R1} is the inlet pressure of the damping hole; p_{R2} is the outlet pressure of the damping hole.

2.3.2. Hydrogen circulation pump

Assuming that the hydrogen temperature after passing through the circulation pump remains unchanged, the relationship between the mass flow rate of the hydrogen, the rotation speed of the circulation pump and the pressure ratio can be obtained through the map shown in Fig. 2. Since the pump outlet pressure and hydrogen flow are corresponding, as long as the pump speed is input, the pump output pressure and flow can be obtained by looking up the table. The calculation of power consumption of circulating pump can be obtained from the adiabatic compression process of ideal gas, and the adiabatic compression process meets the following equation [38]:

$$p \cdot v^k = p_t \cdot v_t^k \quad (2)$$

where, p_t is the pressure of the compressed gas; v_t is the volume specific volume of the corresponding gas; k is the adiabatic index; p and v is the pressure and volume in the compression process.

2.3.3. Air compressor

The dynamic equation of air compressor is [39]:

Table 1
Structural parameters of the purging valve.

Parameter name	Value
Driving voltage	12 V
Coil resistance	14Ω
Viscous damping coefficient	0.2 N.m.s ⁻¹
Initial air gap	2 mm
Spool travel	1.5 mm
Coil inductance	46 mH
Seat half cone angle	45°
Magnetic permeability	1.5 H.m ⁻¹
Armature quality	8 g
Spring stiffness	1.2 N/mm
Preload force of return spring	10 N
Armature inner diameter	4.5 mm

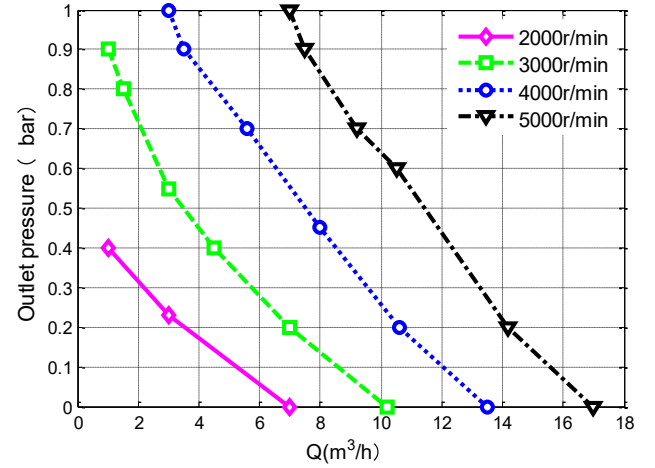


Fig. 2. Characteristic curves of hydrogen circulating pump.

$$\frac{d\omega_m}{dt} = \frac{1}{J_{total}} \cdot (T_m - f \cdot \omega_m - T_{cp}) \quad (3)$$

where, ω_m is the angular speed of the motor; J_{total} is the moment of inertia; T_m is the driving torque of the motor; f is the friction coefficient; T_{cp} is the resistance torque of the air compressor pump head.

The selection of air compressor needs to match with the stack, and the characteristics of air compressor are generally represented by the map of speed flow outlet pressure, as shown in Fig. 3. When the discharge pressure is higher, the force required for compression is greater, and the compressor pressure is inversely proportional to the flow rate at the same speed.

2.3.4. Cathode cavity model

The outlet flow rate of the cathode exhaust pipe is controlled by the electronic three-way throttle valve, and the gas flow through the throttle valve is:

$$W_{rm,out} = \begin{cases} \frac{C_D A_t p_{rm}}{\sqrt{R \cdot T_{rm}}} \pi_t^{\frac{1}{k}} \sqrt{M_{rm} \frac{2k}{k-1} \left(1 - \pi_t^{\frac{k-1}{k}}\right)}, & \pi_t > \left(\frac{2}{k+1}\right)^{\frac{k}{k-1}} \\ \frac{C_D A_t p_{rm}}{\sqrt{R \cdot T_{rm}}} (k M_{rm})^{\frac{1}{2}} \left(\frac{2}{k+1}\right)^{\frac{k+1}{2(k-1)}}, & \pi_t \leq \left(\frac{2}{k+1}\right)^{\frac{k}{k-1}} \end{cases} \quad (4)$$

where, C_D is the mass correction coefficient; A_t is the flow area; p_{rm} is the gas pressure of the exhaust pipe; π_t is the pressure ratio.

According to the conservation of mass, the partial pressure of oxygen and the partial pressure of water vapor inside the cathode cavity can be calculated separately:

$$p_{Ca,O_2} = C_{O_2,ca} \cdot R \cdot T_{fc} \quad (5)$$

$$p_{Ca,H_2O} = C_{vap,ca} \cdot R \cdot T_{fc} \quad (6)$$

$$C_{vap,ca} = \begin{cases} C_{H_2O,ca} & C_{H_2O,ca} < C_{sat,vap}(T_{fc}) \\ C_{sat,vap}(T_{fc}) & C_{H_2O,ca} \geq C_{sat,vap}(T_{fc}) \end{cases} \quad (7)$$

where, T_{fc} is the working temperature of the stack; $C_{vap,ca}$ is the water vapor concentration; $C_{sat,vap}(T_{fc})$ is the saturated water vapor

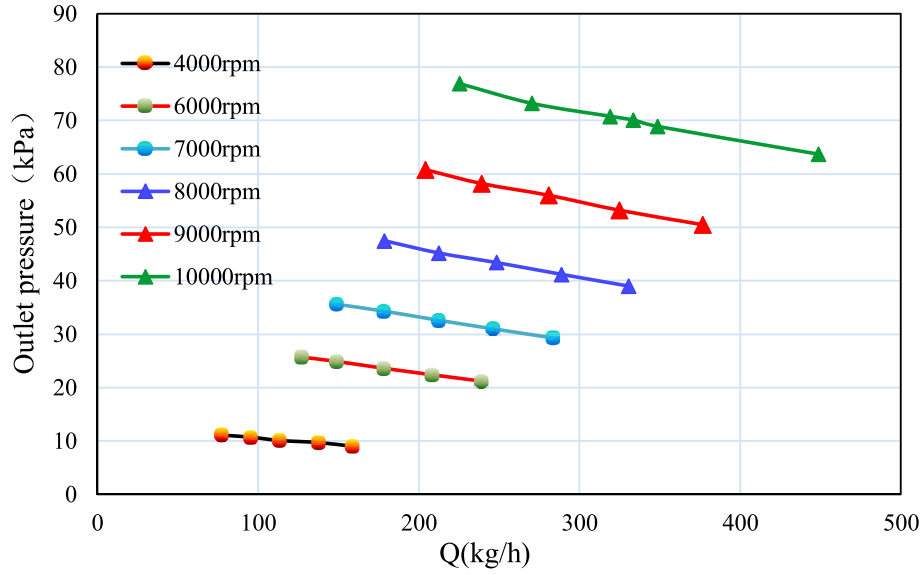


Fig. 3. Performance curves of air compressor.

concentration at this temperature.

In the calculation of the cathode cavity model, the permeation of nitrogen between the anode and cathode is not considered, so the partial pressure of nitrogen is calculated as follows:

$$\frac{dp_{ca-N_2}}{dt} = \frac{R \cdot T_{fc}}{V_{ca} M_{N_2}} (W_{N_2-ca-in} - W_{N_2-ca-out}) \quad (8)$$

where, V_{ca} is the volume of the cathode lumped volume; $W_{N_2-ca-in}$ is the mass flow rate of nitrogen at the cathode inlet; $W_{N_2-ca-out}$ is the mass flow rate of nitrogen at the cathode outlet; M_{N_2} is the molar mass of nitrogen.

The total pressure of the cathode flow channel is:

$$p_{ca} = p_{ca-O_2} + p_{ca-N_2} + p_{ca-H_2O} \quad (9)$$

The mole fractions of nitrogen, hydrogen and water vapor are:

$$x_{O_2} = \frac{p_{ca-O_2}}{p_{ca}} \quad (10)$$

$$x_{N_2} = \frac{p_{ca-N_2}}{p_{ca}} \quad (11)$$

$$x_{H_2O} = \frac{p_{ca-H_2O}}{p_{ca}} \quad (12)$$

Based on this, the mass fraction and flow rate of nitrogen can be calculated:

$$y_{N_2} = \frac{x_{N_2} \cdot M_{N_2}}{x_{O_2} \cdot M_{O_2} + x_{N_2} \cdot M_{N_2} + x_{H_2O} \cdot M_{H_2O}} \quad (13)$$

$$W_{N_2-ca-out} = y_{N_2} \cdot W_{ca-out} \quad (14)$$

liquid two-phase mass transfer, oxygen diffusion, and electrochemical kinetics (Fig. 4). By solving the concentration of material components at the above-mentioned key positions, the distribution rule of the concentration of material components along the direction of proton conduction can be approximated. Combined with the relationship between the concentration of the material component in the flow channel and the concentration of the material component of the peripheral system, a fuel cell engine model based on a double cycle is established for the simulation and control of the actual system.

The hydrogen pressure in the anode channel can be expressed as:

$$\frac{dp_{an-H_2}}{dt} = \frac{R_M T_{an}}{V_{an} M_{H_2}} (W_{H_2-in} - W_{H_2-out} - W_{H_2-con}) \quad (15)$$

where, V_{an} is the anode lumped volume; W_{H_2-out} is the flow rate of hydrogen discharged from the anode; W_{H_2-con} is the rate of hydrogen consumption by the stack.

$$W_{H_2-con} = \frac{M_{H_2} I_{FC} N_{cell}}{2F} \quad (16)$$

The actual fuel cell voltage can be expressed as the theoretical voltage minus the voltage loss due to polarization, as follows:

$$\left\{ \begin{aligned} V(I) &= N_{cell} \cdot (V_{rev} - V_{act,an} - V_{act,ca} - V_{ohmic} - V_{conc,an} - V_{conc,ca}) \\ V_{act,an} + V_{act,ca} &= \frac{R_M T_{FC}}{nF\alpha} \ln\left(\frac{i}{i_0}\right) \Big|_{an} + \frac{R_M T_{FC}}{nF\alpha} \ln\left(\frac{i}{i_0}\right) \Big|_{ca} \\ V_{ohmic} &= I_{FC} \cdot R_{oh} = i \cdot \int_0^{\delta_{men}} \frac{dz}{\sigma(\lambda(z))} \\ v_{conc} &= c \ln\left(\frac{i_L}{i_L - i}\right) \end{aligned} \right. \quad (17)$$

where, R_{oh} is the membrane resistance of a single fuel cell; i is the current density; δ_{men} is the thickness of the proton exchange membrane; σ is the conductivity of the proton exchange membrane.

2.3.5. Stack model

Because the transfer and distribution of multiple substances are involved in the fuel cell, which is relatively complex, this paper will simplify the mechanism model in MATLAB/Simulink. Simulation modeling involves processes such as membrane water transfer, gas-

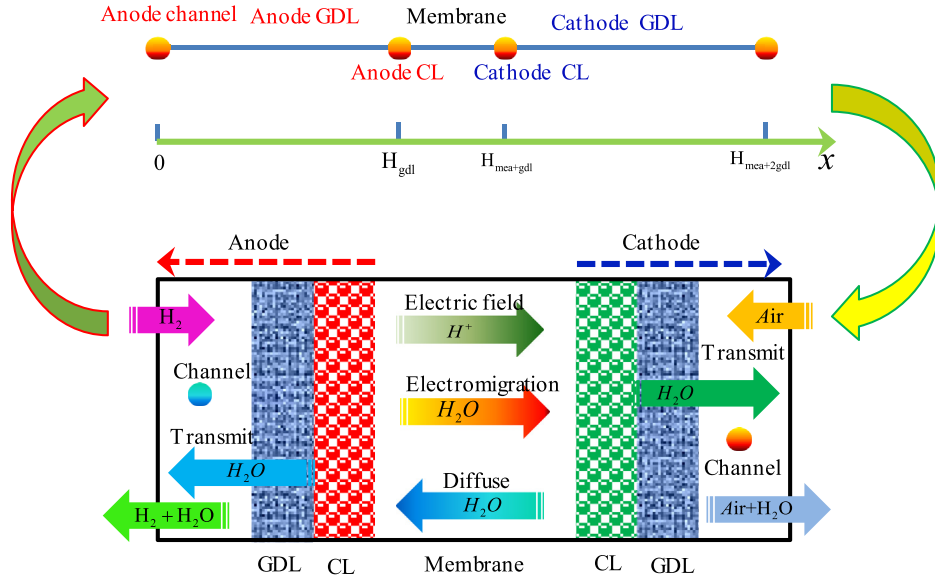


Fig. 4. Schematic diagram of one-dimensional mechanism model.

The relationship between the water content on the anode side $\lambda(0)$ and cathode side $\lambda(\delta_{men})$ of the film and the water vapor activity $a_{vap,an}$ on the surface is as follows:

$$\begin{cases} \lambda(0) = 0.043 + 17.18a_{vap,an} - 39.85a_{vap,an}^2 + 36a_{vap,an}^3 \\ \lambda(\delta_{men}) = 0.043 + 17.18a_{vap,ca} - 39.85a_{vap,ca}^2 + 36a_{vap,ca}^3 \end{cases} \quad (18)$$

Due to the “proton drag” effect, the number of water molecules accompanying each proton is:

$$n_{drag} = n_{drag}^{sat} \frac{\lambda}{22} \quad (19)$$

with the proton from anode to cathode, the molar flow density of water is as follows:

$$J_{H_2O,drag} = n_{drag} \frac{i}{F} \quad (20)$$

flow, so the transport equation for liquid water is:

$$\frac{\partial(\epsilon \rho_l s)}{\partial t} + \nabla \cdot (\rho_l \vec{V} s) = r_w \quad (22)$$

where, ϵ is the porosity, and r_w is the condensation rate.

In the gas diffusion layer (GDL) and the catalyst layer (CL), the phase transformation equation of water is:

$$\begin{cases} \frac{\partial(\epsilon \rho_l s)}{\partial t} + \nabla \cdot \left(\frac{\rho_l K s^3}{\mu_l} \frac{dP_c}{ds} \nabla s \right) = r_w \\ r_w = C_r \cdot \max \left\{ (1-s) \frac{P_{wv} - P_{sat}}{RT} M_{H_2O}, -s \rho_l \right\} \end{cases} \quad (23)$$

where: K is the determining permeability; μ_l is the viscosity of liquid water; P_c is the capillary pressure; C_r is constant condensation rate; P_{wv} is local water vapor pressure; P_{sat} is local saturated steam pressure.

$$\log_{10} P_{sat} = -2.1794 + 0.02953(T - 273.2) - 9.1837 \times 10^{-5}(T - 273.2)^2 + 1.4454 \times 10^{-7}(T - 273.2)^3 \quad (24)$$

The molar flow density of water from cathode to anode is:

$$J_{H_2O,diff} = \frac{\rho_{dry}}{M_{men}} D_\lambda(\lambda) \frac{d\lambda}{dz} \quad (21)$$

where, ρ_{dry} is the dry density of the proton exchange membrane; M_{men} is the molar mass of the proton exchange membrane; D_λ is the diffusion coefficient of water in the proton exchange membrane.

The mode of liquid water transport is different at different positions in the fuel cell, and the liquid water transport in the flow channel, catalyst layer and gas diffusion layer is affected by the gas

Concentration diffusion calculation equation:

$$J_w^{diff} = \frac{\rho_m}{M_m} M_{H_2O} D_l \nabla \lambda \quad (25)$$

where: ρ_m is the density of the dry membrane; M_m is the equivalent of the dry membrane.

Water diffusion coefficient in the membrane:

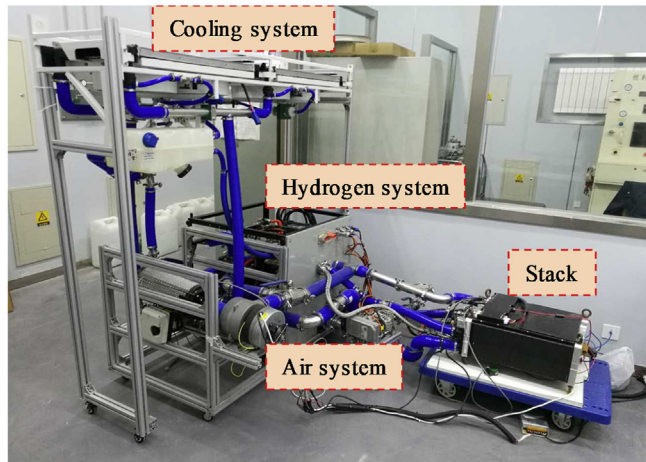
Table 2
Physical and chemical parameters of PEMFC model.

Parameter name	Value	Parameter name	Value
Open circuit voltage	1 V	Porosity of diffusion layer	0.5
Anode reference current density	10000 A/m ²	Diffusion layer density	2719 kg/m ³
Conductivity of diffusion layer	5000	Viscosity resistance coefficient of GDL	10 ¹²
Anode concentration index	0.5	Porosity of catalytic layer	0.5
Anode exchange coefficient	2	Catalytic layer conductivity	5000 S/m
Cathode concentration index	1	Viscosity resistance coefficient of CL	10 ¹²
Cathode exchange coefficient	2	Specific surface area of CL	200,000
Proton transmission impedance	0.0003Ω	Thickness of membrane	0.178 mm
Hydrogen diffusion coefficient	0.00011	Density of membrane	1980 kg/m ³
Oxygen diffusion coefficient	0.000032	Molar mass of membrane	1100 kg/Kmol
Water vapor diffusion coefficient	0.0000735	Conductivity of the membrane	10 ⁻⁶ S/m
Bipolar plate density	2719 kg/m ³	Bipolar plate conductivity	10 ⁶ S/m

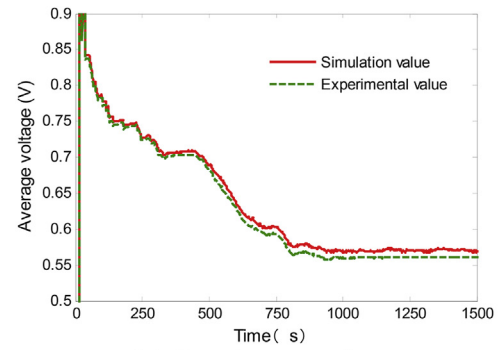
$$D_l = f(\lambda) e^{2416 \left(\frac{1}{303} - \frac{1}{T} \right)} \quad (26)$$

$$f(\lambda) = \begin{cases} 10^{-10}, \lambda < 2 \\ 10^{-10} [1 + 2(\lambda - 2)], 2 \leq \lambda \leq 3 \\ 10^{-10} [3 - 1.67(\lambda - 3)], 3 < \lambda < 4.5 \\ 1.25 \times 10^{-10}, \lambda \geq 4.5 \end{cases} \quad (27)$$

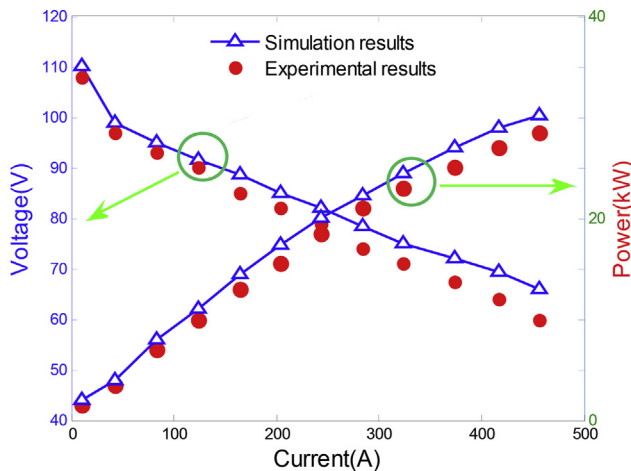
The stack model contains the mass transfer process of gases such as oxygen, water vapor or liquid water, as well as the nonlinear equations for the calculation of the stack voltage, so many parameters are involved. However, since most parameters are not measurable, the structural parameters and material parameters in this article basically refer to the data provided by the manufacturer and the literature [40]. Some important parameters required for the stack simulation model are shown in Table 2.



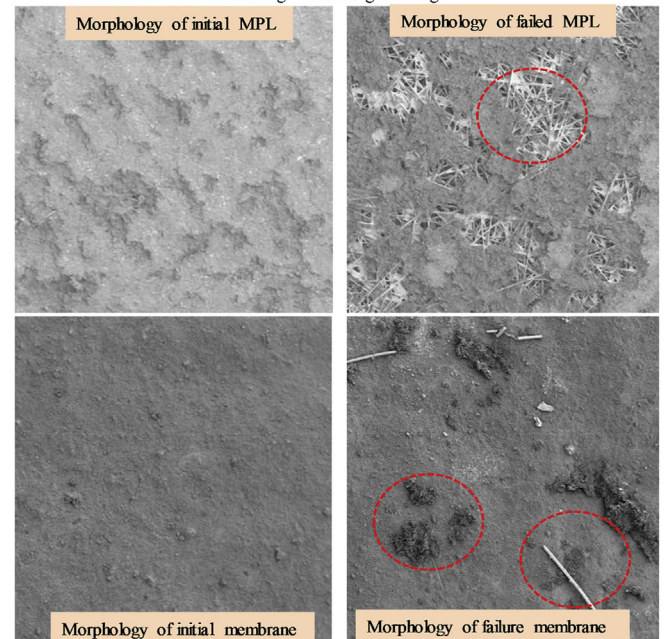
(a) PEMFC engine



(c) Change in average voltage



(b) Validation of proposed model



(d) Microstructure of MPL and membrane surface under dry conditions

Fig. 5. Simulation and experimental results.

3. Results and discussion

3.1. Model verification

In order to verify the validity of the simulation model, the simulation and experimental values of PEMFC engine without cathode recirculation were compared. As showed in Fig. 5a, the fuel cell system test bench includes hydrogen system, air system, and cooling system and control system. The air system includes air compressor, mass flowmeter and back pressure valve; the hydrogen system includes hydrogen cylinder, purge valve, high-pressure regulating valve, mass flowmeter, low-pressure regulating valve, hydrogen circulation pump, etc.; the cooling system includes water pump, water tank, radiator, etc.; the power output system is connected with single cell voltage inspection, electronic load, etc. The external conditions of the experimental process are as follows: neither anode nor cathode use external humidification; the excess ratios of cathode and anode are 2 and 1.5, respectively; the ambient temperature and the working temperature of the stack are 25 °C and 60 °C, respectively. The specific parameters are shown in Table 3. The comparison between the simulated polarization curve obtained in this paper and the experimental results is shown in Fig. 5b. It is found that they have good consistency with a maximum error of only 5%, indicating that the one-dimensional PEMFC engine model used herein can be considered as valid. In addition, we also found that the simulated value was slightly higher than the experimental value, and the larger the output power was, the greater the error would be. The foremost reason is that the cathode side outlet of the stack is prone to flooding, while the simulation model cannot accurately simulate the process, and can only ignore the impact of the cathode outlet flooding. Due to the fact that the internal water state parameters of the stack, such as the water content of the membrane, catalyst layer, and diffusion layer, cannot be measured, the model has a process of initial value adjustment and correction, as shown in Fig. 5c. The initial value of the relative error of the calculated average single cell voltage is low, and the error gradually increases after that due to the complicated working environment under high power. From the SEM results of the MPL layer and membrane of Fig. 5d, it can be seen that the carbon black carrier is corroded and peeled off due to the membrane drying after long-term operation without humidification. These impurities not only adhere to the surface of the carbon paper and block the carbon paper, which will inevitably reduce the permeability and hydrophobicity of the diffusion layer, resulting in a decrease in the performance of the stack. Corrosion of the catalytic layer and carbon black carrier leads to agglomeration and sintering of the catalyst particles. The membrane suffered serious mechanical performance damage, and the membrane surface was uneven, with black impurities and broken fibers on the carbon paper. These result in the impedance of the film exceeding 400 mΩ cm², or even reaching 600 mΩ cm², and the effective mass transfer area to become smaller, which ultimately leads to the degradation of the stack performance.

Table 3
Main performance parameters of the stack.

Stack index	Value
Rated power	30 kW
Active area	350 cm ²
Number of fuel cells	100 cells
Voltage range	70–120 V
Current range	0–500 A

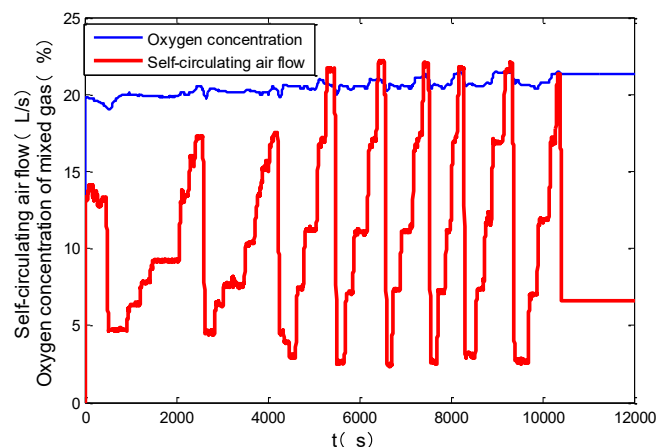


Fig. 6. Effect of air self-circulation on oxygen concentration.

3.2. Self-circulating humidification performance

In order to maintain the water balance inside the fuel cell, the amount of water obtained by the anode must be equal to the net migration of water to the cathode. Therefore, measures to increase the water vapor permeation rate from the cathode to the anode by increasing the water concentration gradient on both sides of the membrane to compensate for the reverse electroosmosis loss through humidifying the cathode air intake are needed. Since the water produced by the fuel cell during operation will lead to a large amount of high-temperature saturated water vapor in the cathode exhaust gas, using it to directly moisten the inlet gas can not only realize the self-humidification of the fuel cell, but also increases the stack's intake pressure.

Fig. 6 shows the simulation results of the oxygen concentration of the mixed gas at the cathode inlet as a function of the air circulation volume, and the amount of recirculated air was changed by adjusting the opening of the electronic three-way valve. Since most of the oxygen will be consumed after the air passes through the stack, the oxygen concentration at the stack outlet will be lower. The air at the outlet of the stack is pumped to the inlet of the stack because of the strong suction of the compressor and mixed with fresh air will also reduce the oxygen concentration of the mixed gas. Although the oxygen concentration at the inlet of the stack decreases with the increase of the circulating air flow, the change in the oxygen concentration is very small, indicating that it will not affect the oxygen partial pressure of the mixed gas. As shown in the simulation results shown in Fig. 7, the humidity of the mixed gas at the cathode inlet of the stack gradually increases as the current density increases from 0.2 A/cm² to 0.95 A/cm² due to the inflow of high-heat, high-humidity exhaust gas. It is worth noting that the maximum humidity of the mixed gas in the whole process does not exceed 60% under the passive operation of the cathode self-circulation system. Moreover, during the loading process, there is a certain delay in the change in humidity, possibly because there is a certain distance between the exhaust port of the stack cathode and the mixing chamber in front of the compressor, resulting in a certain lag in the change in gas flow. However, when the stack enters the steady state, it quickly returns to the steady state value, so the stability of the relative humidity at the stack inlet can be guaranteed in the process of loading and unloading. It is also observed that there is a certain inflection point in the increasing trend of the humidity of the mixture at the current density of 0.65 A/cm², and the increasing rate after it exceeds 0.65 A/cm² obviously becomes larger. This is because the greater the current density, the

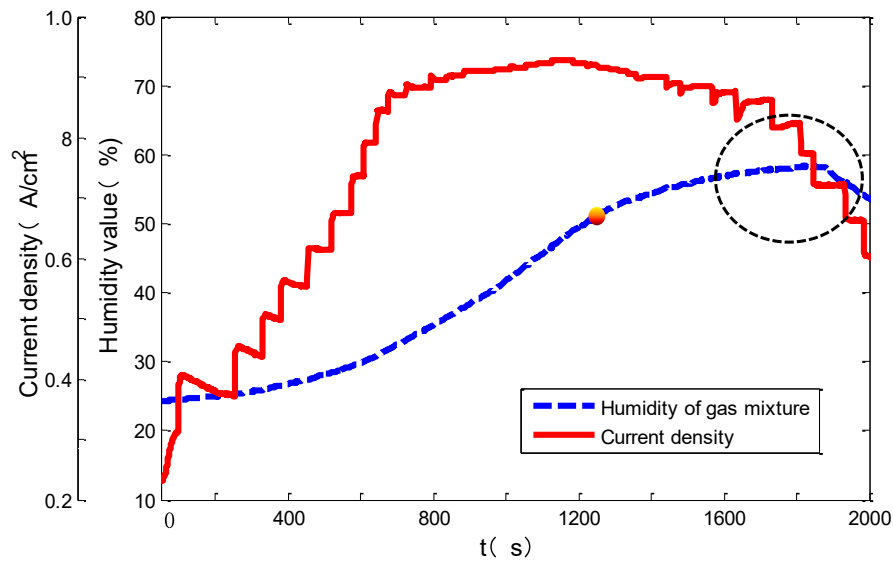
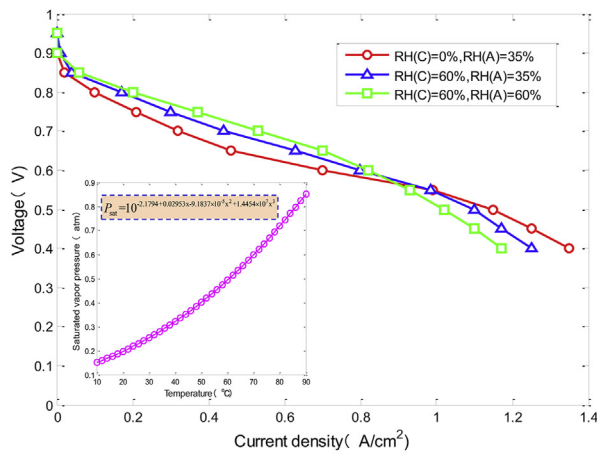
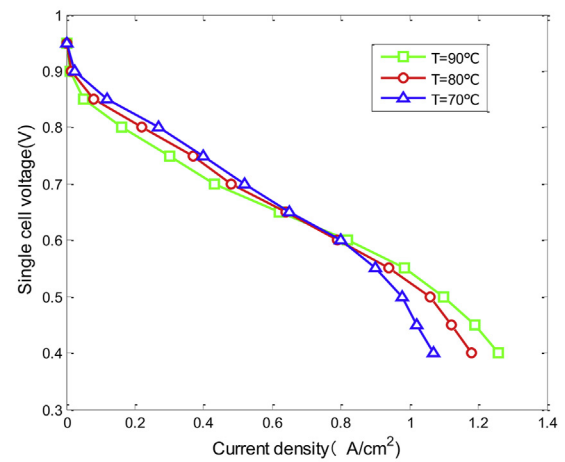


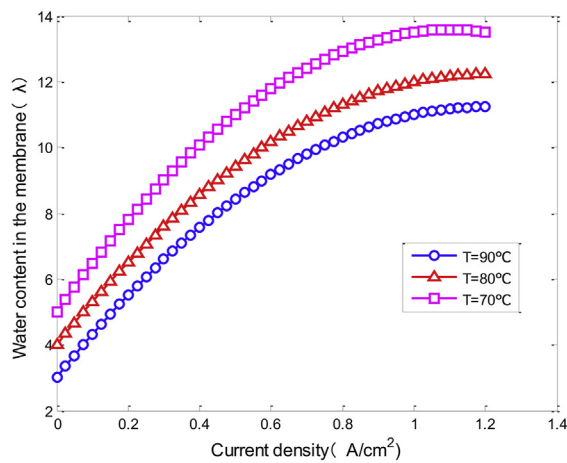
Fig. 7. Humidity change of mixed gas under different conditions.



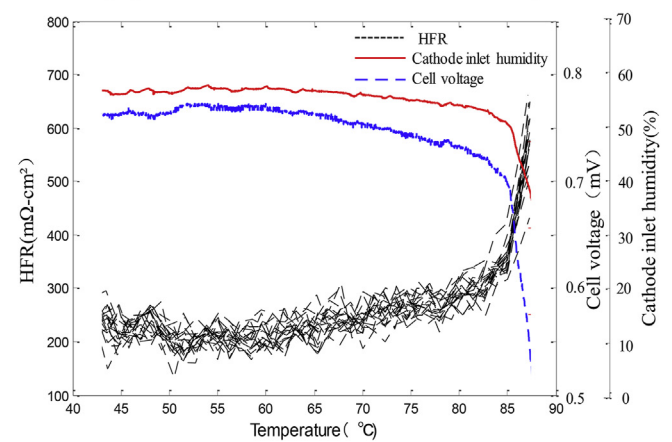
(a) The influence of temperature on gas humidity and fuel cell voltage



(c) Polarization curve of cathode self-circulation system



(b) Effect of temperature on average water content in membrane



(d) Influence of temperature on the voltage and humidity

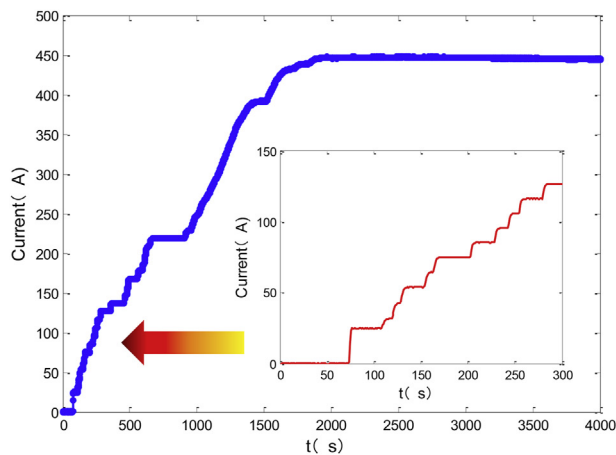
Fig. 8. Effect of temperature on the performance of fuel cells.

more water is produced from the cathode and the greater the humidity of the exhaust gas.

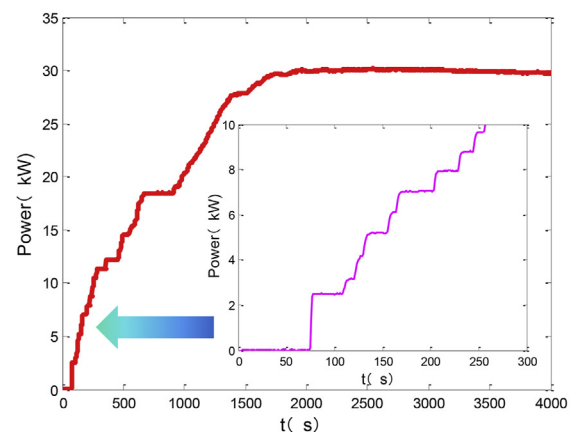
As we all know, temperature is an important parameter in the study of thermal management of PEMFC, that is, too low temperature will lead to the performance degradation of fuel cell, while high temperature will lead to too much membrane water loss and normal operation. Therefore, studying the influence of temperature on fuel cell performance is of practical significance for obtaining the optimal working conditions of fuel cell engines. The fuel cell operating conditions required for the simulation are: temperature and pressure are 60 °C and 1 atm, respectively; the excess coefficients of the cathode and anode are 2 and 1.5, respectively. From the curve of water vapor saturation pressure versus temperature given in Fig. 8a, it can be known that the saturated vapor pressure increases exponentially with the rise of temperature, and at a higher operating temperature, even a temperature difference of several degrees can significantly increase the water content of the gas. As the current density increases and the water production of the cathode catalyst layer increases, the low-humidity gas is more conducive to gas diffusion, and thus has a higher output voltage. Fig. 8b is the simulation result of the effect of temperature on the average water content in the membrane. The higher the temperature, the more the water in the membrane evaporates, resulting in worse water and conditions in the proton exchange membrane. As

the current density increases, the amount of water produced in the cathode catalytic layer increases, and the more liquid water formed in the cathode diffusion layer. The high working temperature is conducive to the evaporation of water vapor, and the flooding of the diffusion layer and the catalytic layer is reduced. Furthermore, the recycling of cathode exhaust gas increases the humidity of the inlet, which has better output performance, as shown in Fig. 8c. In addition, it can be seen from Fig. 8d that the average cell voltage has a good correspondence with high-frequency impedance, that is, the voltage is high at low impedance, the voltage is low at high impedance, and there is almost no delay, indicating that high-frequency impedance is important for fuel cell systems. When the membrane is too dry, the average single cell voltage decreases from 0.78 V to 0.7V during the rapid rise of the membrane impedance from 300 to 600 $\text{m}\Omega\text{ cm}^2$, indicating that the performance degradation is about 10%. Because the average single cell voltage almost declines in a straight line after 85 °C, it can be judged that the voltage has been unstable at this time. Therefore, when the high frequency impedance is in the range of 200–400 $\text{m}\Omega\text{ cm}^2$, the stack is in an ideal working state.

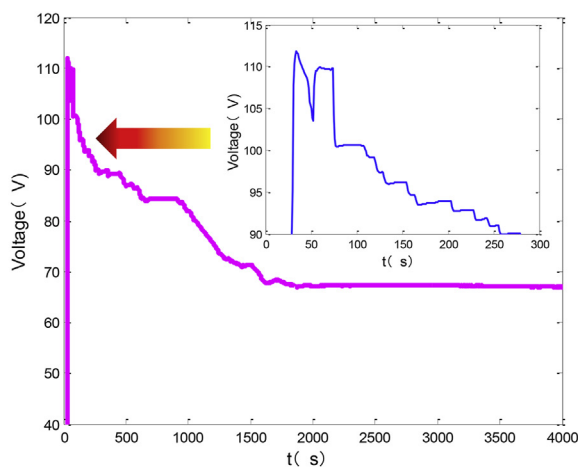
The dynamic and steady-state performance of PEMFC engine based on air self-circulation humidification system is shown in Fig. 9, where the simulation conditions are: $T = 70\text{ }^\circ\text{C}$, cathode gas excess factor 2 and anode gas excess factor 1.5. The current is



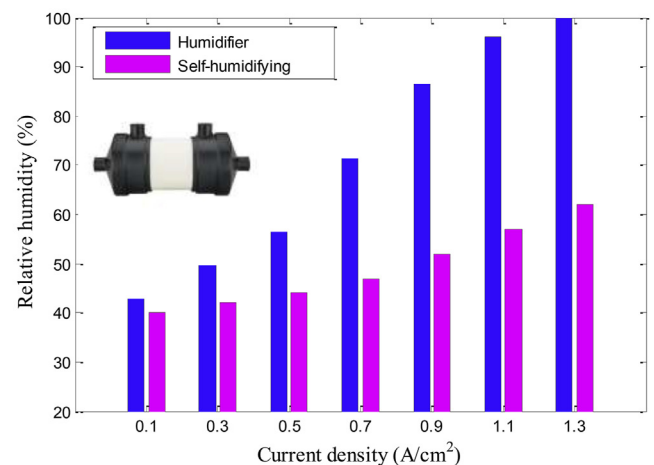
(a) Changes in current



(c) Changes in stack power



(b) Dynamic and steady state changes of stack voltage



(d) Comparison of the effects of two humidification methods

Fig. 9. Performance of the stack with the air self-circulation system.

gradually loaded by step, and it takes about 20 s for the current to reach the stable state of the set value as shown in Fig. 9a. The difference is that the larger the current, the shorter the time required. The voltage variation trend of the stack is just opposite to that of the current, as shown in Fig. 9b. When the stack is powered on, the stack will be purged first, so the voltage will rise. After the purging, the compressor stops running, and the inlet and outlet of the stack are closed, resulting in the gradual consumption of internal oxygen. After the stack starts running, the voltage drops suddenly for about 50s. After that, the compressor works with 20 A current, so the voltage rises again. As the current increases, it takes about 30s for the voltage to reach a stable state. When the current is loaded to 450 A, the power of the stack reaches 30 kW, and can maintain a relatively stable state during long-term operation, as shown in Fig. 9c. After adopting the cathode recycling system, the voltage fluctuation of the stack is small and the output power is stable, indicating that it has good reliability. The self-humidifying capacity based on the cathode and anode double circulation system is shown in Fig. 9d. At low current density, the humidification capacity of the two modes of self-circulation and humidifier is almost equal, but the greater the current density, the better the effect of the humidifier can reach 100%, while the maximum value of self-circulation can only reach 60%. However, as the thickness of the electrolyte membrane is reduced, moisture will move from a place

with high humidity (downstream of air flow) to a place with low humidity (upstream of hydrogen flow). As the air pole will generate water with the generation reaction of fuel cell, the higher the humidity of air is downstream. The humidity difference between the downstream of the air flow and the upstream of the hydrogen flow increases the humidity of the hydrogen upstream and the humidity of the hydrogen also increases downstream of the flow. The air recirculation system increases the relative humidity at the cathode inlet and the water content of the membrane, thereby improving the output performance of the stack. Therefore, not only is the durability problem caused by high potential at a small current density alleviated, but the performance and stability of the stack can be improved.

3.3. Cathode and anode control strategies

Based on the model of the dual cycle fuel cell engine, the influence of different cathode recycling rate on the output performance of the stack without external humidifier is studied, and the map of the application scope of the self-humidification function of the dual cycle system is obtained. When the current is gradually loaded, the opening and closing frequency of the purging valve increases gradually, but the opening time under different working conditions is constant, about 10 s, as shown in Fig. 10a. The purpose

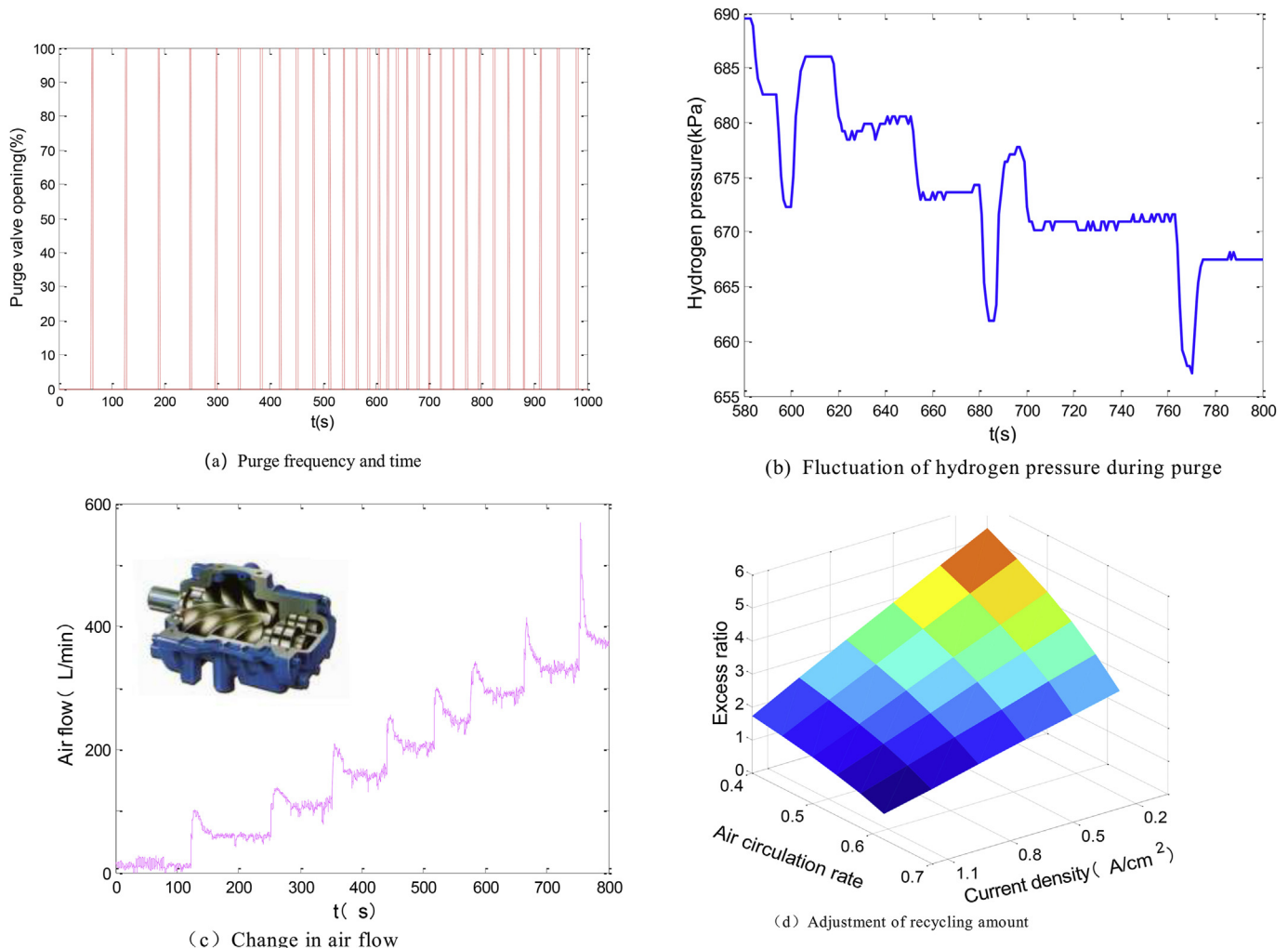


Fig. 10. Control of double circulation self-humidifying system.

of purging is to remove the generated flying and nitrogen penetrating from the cathode to the anode, so as to improve the purity of hydrogen. Due to the influence of the opening and closing of the purge valve, the hydrogen pressure will be reduced by about 10 kPa each time, as shown in Fig. 10b. Fig. 10c shows the dynamic change characteristics of air compressor flow rate with load current. When the load is low, it is difficult for the flow rate of the air compressor to stabilize quickly, and the overshoot phenomenon occurs. As the current increases, the dynamic performance of the air compressor gradually gets better, the overshoot is smaller, and it can quickly reach a stable state. It was found that increasing the current density loading process time while reducing the loading amplitude is beneficial to improve the stability of the battery and extend the life of the stack. By controlling the opening of the electronic three-way valve at the cathode outlet to adjust the self-circulation rate of air (the proportion of circulation volume to the mixed gas), it is necessary to cooperate with the working state (current density) of the stack and the required air supply excess ratio to achieve self-humidification, as shown in Fig. 10d.

4. Conclusion

In this study, the cathode air self-circulation system of fuel cell engine is designed, and the performance of PEMFC engine is studied from three aspects: one-dimensional modeling, bench test and simulation model. Depending on the high flow speed of tail gas and the regulation of electronic three-way valve, the self-circulation humidification function without circulation pump is realized. The following conclusions can be drawn:

- (1) The polarization curve and output power predicted by the calibrated PEMFC system coupling simulation model shows good agreement with experimental data for constant-current operations. In addition, the average single cell voltage has good dynamic characteristics and its value is gradually reduced for long-time operation of 30 kW.
- (2) The recirculation system can significantly increase the relative humidity of the stack inlet and the water content of the membrane, but has a smaller effect on the oxygen concentration. At high current densities, increasing the operating temperature can reduce the liquid water saturation on the cathode side and prevent fuel cell flooding.
- (3) Cathode recirculation has a better dynamic effect of humidification, which is more suitable for short-term low inlet humidity working conditions. With the change of the opening of the electronic three-way valve, the humidity at the inlet can reach 60%, which solves the problem of membrane drying at the inlet due to the lack of external humidifier with low current density.

The results of the study are of great significance for optimizing the operational reliability of fuel cell and improving water management capability of PEMFC system. Based on the work of this paper, we can continue to carry out the following research work: carry out experiments on the impact of recycling system on the durability of the stack; the designed control algorithm fails to carry out algorithm verification on the fuel cell engine, which can be applied to the fuel cell engine later, and optimize the control strategy of the fuel cell engine.

Author statement

Qinguo Zhang: Writing - original draft, Write and revise papers. Zheming Tong: Experimental guidance. Shuiguang Tong: Research guidance. Zhewu Cheng: Modeling and data processing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (52075481), Zhejiang Provincial Natural Science Foundation (LR19E050002, LQ21E060009), Zhejiang Province Key Science and Technology Project (2018C01020, 2018C01060, 2019C01057).

Nomenclature

a	Water vapor activity
A	Area, flow cross-sectional area(m^2)
C	Substance concentration (mol/m^3)
d	Width or aperture (m)
D	Gas diffusion coefficient ($\text{m}^2 \cdot \text{s}^{-1}$)
f	Friction factor
F	Faraday constant (96,487 C/mol)
G	Gibbs Free Energy (J/kg)
i	Current density (A/m^2)
I	Current (A)
K	Permeability (m^2)
k	Permeability coefficient($10^{-13} \text{m}^2/\text{s}$)
m	Mass (kg)
M	Molecular molar mass ($\text{kg} \cdot \text{mol}^{-1}$)
s	Liquid water saturation
S	Phase change rate ($\text{mol} \cdot \text{m}^{-3} \cdot \text{s}^{-1}$)
Q	Mass flow rate (kg/m^3)
R	Gas constant ($\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$)
P	Gas pressure (Pa)
T	Temperature ($^{\circ}\text{C}$)
U	Voltage (V)
V	Velocity (m/s), volume (m^3)
ε	Porosity
θ	Contact angle ($^{\circ}$)
κ	Membrane conductivity($\Omega^{-1} \cdot \text{m}^{-1}$)
ρ	Density ($\text{kg} \cdot \text{m}^{-3}$)
ν	Viscosity (m^2/s)

Abbreviation

An	Anode
Ca	Cathode
CL	Catalytic layer
GDL	Gas diffusion layer
Sim	Simulation value
Exp	Experimental value
Sat	Saturation
MEA	Membrane electrode assembly
Vap	Vapor
Ch	Channel

Superscripts and subscripts

Out	Outlet
In	Inlet
l	liquid
g	Gas
ex	external
to	total
Mem	Membrane
H_2	In Hydrogen Inlet

Fc	Fuel cell
Cp	Compressor
Oh	Ohmic
h	hour
W	Water
P	Power
Gc	Gas channel

References

- [1] Zhenwu Cheng, Shuiguang Tong, Zheming Tong, Bi-directional nozzle control of multistage radial-inflow turbine for optimal part-load operation of compressed air energy storage[J], *Energy Convers. Manag.* 181 (1) (2019) 485–500.
- [2] Shuiguang Tong, Zhenwu Cheng, Feiyun Cong, Zheming Tong, Yidong Zhang, Developing a grid-connected power optimization strategy for the integration of wind power with low-temperature adiabatic compressed air energy storage [J], *Renew. Energy* 125 (2018) 73–86.
- [3] Yubai Li, Zhifu Zhou, Xianglei Liu, Weitao Wu, Modeling of PEM fuel cell with thin MEA under low humidity operating condition [J], *Appl. Energy* 242 (15) (2019) 1513–1527.
- [4] Xiaohui Yan, Cheng Guan, Yao Zhang, Flow field design with 3D geometry for proton exchange membrane fuel cells[J], *Appl. Therm. Eng.* 147 (25) (2019) 1107–1114.
- [5] Chunhua Min, He Jing, Kun Wang, Liyao Xie, Xuguang Yang, A comprehensive analysis of secondary flow effects on the performance of PEMFCs with modified serpentine flow fields [J], *Energy Convers. Manag.* 180 (15) (2019) 1217–1224.
- [6] Jun Shen, Zhengkai Tu, Siew Hwa Chan, Enhancement of mass transfer in a proton exchange membrane fuel cell with blockage in the flow channel[J], *Appl. Therm. Eng.* 149 (25) (2019) 1408–1418.
- [7] F.B. Baz, Mahmoud Ahmed Shinichi Oookawara, Enhancing under-rib mass transport in proton exchange membrane fuel cells using new serpentine flow field designs[J], *Int. J. Hydrogen Energy* 44 (58) (2019) 30644–30662.
- [8] Pucheng Pei, Yuehua Li, Huachi Xu, Ziyao Wu, A review on water fault diagnosis of PEMFC associated with the pressure drop[J], *Appl. Energy* 173 (1) (2016) 366–385.
- [9] Pucheng Pei, Huachi Xu, Xia Zeng, Hongshan Zha, Mancun Song, Use of galvanostatic charge method as a membrane electrode assembly diagnostic tool in a fuel cell stack[J], *J. Power Sources* 245 (1) (2014) 175–182.
- [10] Mancun Song, Pucheng Pei, Hongshan Zha, Huachi Xu, Water management of proton exchange membrane fuel cell based on control of hydrogen pressure drop [J], *Journal of Power sources* 267 (1) (2014) 655–663.
- [11] Rupak Banerjee, G. Satish, Kandlikar, Experimental investigation of two-phase flow pressure drop transients in polymer electrolyte membrane fuel cell reactant channels and their impact on the cell performance [J], *J. Power Sources* 268 (5) (2014) 194–203.
- [12] Rupak Banerjee, Valentina Mejia Danielle Howe, G. Satish, Kandlikar, Experimental validation of two-phase pressure drop multiplier as a diagnostic tool for characterizing PEM fuel cell performance[J], *Int. J. Hydrogen Energy* 39 (31) (2014) 17791–17801.
- [13] Yifan Xu, Diankai Qiu, Peiyun Yi, Shuhuai Lan, Linfa Peng, An integrated model of the water transport in nonuniform compressed gas diffusion layers for PEMFC[J], *Int. J. Hydrogen Energy* 44 (26) (2019) 13777–13785.
- [14] Eun Jung Choi, Jin Young Park, Min Soo Kim, A comparison of temperature distribution in PEMFC with single-phase water cooling and two-phase HFE-7100 cooling methods by numerical study[J], *Int. J. Hydrogen Energy* 43 (29) (2018) 13406–13419.
- [15] Yonghua Cai Ben Chen, Zhengkai Tub Jun Shen, Siew Hwa Chan, Performance degradation of a proton exchange membrane fuel cell with dead-ended cathode and anode[J], *Appl. Therm. Eng.* 132 (5) (2018) 80–86.
- [16] Bachir Dadda, Said Abboudi, Rida Zarrit, Heat and mass transfer influence on potential variation in a PEMFC membrane [J], *Int. J. Hydrogen Energy* 39 (27) (2014) 15238–15245.
- [17] Peng Quan, Lai Ming-Chial, Numerical study of water management in the air flow channel of a PEM fuel cell cathode [J], *J. Power Sources* 164 (1) (2007) 222–237.
- [18] Z. Liu, Z. Mao, C. Wang, A two dimensional partial flooding model for PEMFC [J], *J. Power Sources* 158 (2) (2006) 1229–1239.
- [19] Jinyong Kim, Gang Luo, Chao-Yang Wang, Modeling two-phase flow in three-dimensional complex flow-fields of proton exchange membrane fuel cells[J], *J. Power Sources* 365 (15) (2017) 419–429.
- [20] TaoFeng Cao, Hong Lin, Numerical investigation of the coupled water and thermal management in PEM fuel cell[J], *Appl. Energy* 12 (10) (2013) 1115–1125.
- [21] Li Chen, TaoFeng Cao, Numerical investigation of liquid water distribution in the cathode side of proton exchange membrane fuel cell and its effects on cell performance [J], *Int. J. Hydrogen Energy* 37 (1) (2012) 9155–9170.
- [22] Anh Dinh, Biao Zhou, A generalized numerical model for liquid water in a proton exchange membrane fuel with interdigitated design [J], *J. Power Sources* 193 (2) (2009) 665–683.
- [23] WeiMon Yana, Hsin-Sen Chub, Jian-Yao Chen, Transient analysis of water transport in PEM fuel cells[J], *J. Power Sources* 162 (2) (2006) 1147–1156.
- [24] Falin Chen, YuGuang Su, Chyi-Yeou Soong, Transient behavior of water transport in the membrane of a PEM fuel cell[J], *J. Electroanal. Chem.* 566 (1) (2004) 85–93.
- [25] F.N. Büchi, S. Srinivasan, Operating proton exchange membrane fuel cells without external humidification of the reactant gases fundamental aspects [J], *J. Electrochem. Soc.* 144 (8) (1997) 2767–2772.
- [26] M.V. Williams, H.R. Kunz, J.M. Fenton, Operation of Nafion®-based PEM fuel cells with no external humidification: influence of operating conditions and gas diffusion layers [J], *J. Power Sources* 135 (1) (2004) 122–134.
- [27] Z. Qi, A. Kaufman, PEM fuel cell stacks operated under dry-reactant conditions [J], *J. Power Sources* 109 (2) (2002) 469–476.
- [28] M. Watanabe, H. Uchida, Y. Seki, et al., Self-humidifying polymer electrolyte membranes for fuel cells[J], *J. Electrochem. Soc.* 143 (12) (1996) 3847–3852.
- [29] F. Liu, B. Yi, D. Xing, et al., Development of novel self-humidifying composite membranes for fuel cells[J], *J. Power Sources* 124 (1) (2003) 81–89.
- [30] U.H. Jung, K.T. Park, E.H. Park, et al., Improvement of low-humidity performance of PEMFC by addition of hydrophilic SiO₂ particles to catalyst layer[J], *J. Power Sources* 159 (1) (2006) 529–532.
- [31] B.J. Kim, M.S. Kim, Studies on the cathode humidification by exhaust gas recirculation for PEM fuel cell[J], *Int. J. Hydrogen Energy* 37 (5) (2012) 4290–4299.
- [32] Yangbin Shao, Liangfei Xu, Xingwang Zhao, Jianqiu Li, Minggao Ouyang, Comparison of self-humidification effect on polymer electrolyte membrane fuel cell with anodic and cathodic exhaust gas recirculation [J], *Int. J. Hydrogen Energy* 45 (4) (2020) 3108–3122.
- [33] Bohan Wang, Rui Lin, Dengcheng Liu, Ji Xu, Bowen Feng, Investigation of the effect of humidity at both electrode on the performance of PEMFC using orthogonal test method [J], *Int. J. Hydrogen Energy* 26 (26) (2019) 13737–13743, 44.
- [34] S. Rodosik, J.P. Poirot-Crouvezier, Y. Bultel, Impact of humidification by cathode exhaust gases recirculation on a PEMFC system for automotive applications [J], *Int. J. Hydrogen Energy* 44 (25) (2019) 12802–12817.
- [35] Xingwang Zhao, Liangfei Xu, Chuan Fang, Hongliang Jiang, Study on voltage clamping and self-humidification effects of pem fuel cell system with dual recirculation based on orthogonal test method [J], *Int. J. Hydrogen Energy* 43 (33) (16 August 2018) 16268–16278.
- [36] Junmin Hu, Jianqiu Li, Liangfei Xu, Fusen Huang, Minggao Ouyang, Analytical calculation and evaluation of water transport through a proton exchange membrane fuel cell based on a one-dimensional model[J], *Energy* 111 (15) (2016) 869–883.
- [37] Huicui Chen, Biao Liu, Tong Zhang, Pucheng Pei, Influencing sensitivities of critical operating parameters on PEMFC output performance and gas distribution quality under different electrical load conditions [J], *Appl. Energy* 255 (2019) 113849.
- [38] Xingwang Zhao, Liangfei Xu, Chuan Fang, Hongliang Jiang, Study on voltage clamping and self-humidification effects of pem fuel cell system with dual recirculation based on orthogonal test method[J], *Int. J. Hydrogen Energy* 43 (2018) 16268–16278.
- [39] Liangfei Xu, Chuan Fang, Junming Hu, Siliang Cheng, Parameter extraction of polymer electrolyte membrane fuel cell based on quasi-dynamic model and periphery signals[J], *Energy* 122 (1) (2017) 675–690.
- [40] Yujiao Chen, Zheming Tong, Yang Zheng, Holly Samuelson, Leslie Norford, Transfer learning with deep neural networks for model predictive control of HVAC and natural ventilation in smart buildings[J], *J. Clean. Prod.* 254 (2020) 119866.